6.S974/2.S988/1.S980: Games, Learning, and Security Fall 2026

Wargame Deliverable 1

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1 Setting and Model

(a) **Setting:**
This game is set to be a simulated game of maritime control in the East and South China Seas. The game will be a two player game composed of Chinese naval forces and US Naval forces. The areas of contest include the titular waters, with particular focus on the waters surrounding Taiwan, and identified areas of interest contained within the First Island Chain.

(b) **Model:**
In this game, US maritime components seek to dominate the areas of contest and control the waterspace without escalating to war. China seeks to control these waters as well, and establish further control reaching beyond them, also without escalating to war. The areas of interest, and the way "control" is measured within the game will be a function of forces within the sea boundaries stated, with particular weight given to vessels in proximity of a set of important features in the waters, namely Taiwan and politically sensitive islands in the First Island Chain (i.e. the Spratly Islands, Scarborough Shoal).

The available actions to each player will be the ability to allocate forces to an area, then the follow-on actions in that area (i.e. presence, surveillance, attack, retreat). The number of forces available to each player will be initialized, then remain a dynamic variable throughout the game due to constraints and effects of actions.

Information available to each player will be history of the game thus far, predictions of what the other players will do, and any information gained through their own actions.

Resource constraints to each force will be food, fuel, and maintenance.

The system state, will be defined by the player's actions over time. Payoff will be evaluated at every step in sequence of play. Sequence of play will be simultaneous, determined on some time interval for each step in the game, where state will be evaluated. Each player's payoff at the most recent state in the game, represents the true system state. Payoffs are not cumulative and should be decided by current state. The game will be determined to be over when either side escalates to war, or when a threshold payoff, to be determined, is reached by one of the players.

2 Relevant Literature:

(a) *Understanding and Countering China's Maritime Gray Zone Operations — Helmus et al. (2024).* This source describes China's strategies and tactics for asserting influence in the South China Sea and discusses approaches to countering coercive maritime operations. We will use it to develop an action space and resource constraints grounded in real-world considerations. Our game will translate these qualitative descriptions into explicit rules and payoffs so that we can systematically compare strategies.

(b) *Gaming Gray Zone Tactics: Design Considerations for a Structured Strategic Game — Wasser et al. (2019).* This source describes the development of a structured game examining gray-zone competition between Russia and the West in the Balkans. We will draw on its approach to organizing player objectives, available actions, and game rules to represent competition below the threshold of outright war. While Helmus et al. will inform the content of our decision space, Wasser et al. will inform its structure. Our game will adapt these design principles to maritime competition in the South China Sea.

3 Research Questions:

(a) What strategy types, i.e. aggressive, moderate, or conservative, yield the greatest success in denying the opponent control of the South China Sea while limiting escalation(or other region depending on the details of the game)? We will define these types by their willingness to select actions with greater escalation risk and compare them under the same initial conditions and resource constraints. We will measure success through the duration of access denial, resource expenditure, and frequency of escalation. A fixed moderate policy will serve as a simple benchmark.

(b) As a preliminary to the first question, how can real-world strategies outlined in the Helmus, et. al. (2024) be represented as player types and action policies and how do differences in action selection probabilities affect game outcome? Holding the available actions, initial state and information constant, we will represent each policy through different probabilities of selecting feasible actions. We will examine whether small changes in these probabilities lead to gradual or substantial changes in access denial and escalation. This will test whether the conclusions from our first question are robust to how we define aggressive, moderate, and conservative behavior.

4 Plan and Feedback

(a) Prior to Workshop 1, we will produce a Python implementation of our game that steps through 20 iterations of the game. For this iteration, we intend to keep the scope narrower, focusing on implementing a working model, with a narrower choice of options at each iteration, a static information set, and constraints. We will also keep the inventory of vessels to each force as 10, to begin with. A small initial instance of the game could be such that China sends 5 ships to Okinawa and 5 ships to Taiwan. The US sends 5 ships to Scarborough Reef and 5 to Taiwan. Both choices are assumed to be rational in this instance. The payoff China gets is 100 for advancing their area of operations. The payoff the US gets is 50 for occupying Scarborough Reef. If we assume both players get an equal payoff for Taiwan, then we that payoff cancels out when evaluating the state of the game at that point.

For incorporating learning, we intend to implement learning after we gain a more formal understanding of it in class, and can produce results with a fixed information set. Learning will be implemented in such a manner that with every iteration of time, and maybe the execution of some actions (like surveillance) each player is able to infer aspects of the others manner of play and use that in decision calculus. As far as incorporating attacks and defenses, we intend to implement that as a large aspect of the game once our escalation threshold of the game is met. If that happens in the game, state is no longer determined by who has the highest payoff in a situation, and our objective will need to change to be evaluated on who has the most forces still in the fight. Termination of the game will then be determined by a minimum threshold or end score once a predetermined number of iterations have completed.[8pt] The main modeling choices that we would like feedback on are: how do we implement where to send each vessel in each players Fleet at each turn of the game? Assuming it is a Mixed Integar Programming problem, how do we also incorporate data into that? We will most likely request more feedback throughout the semester, but this is our most pertinent modeling question to build our first iteration of the game prior to workshop one.

5 References:

(a) Helmus, T. C., Romita Grocholski, K., Liggett, T., Rhoades, A. L., Savitz, S., and Palmer, K. (2024). Understanding and countering China's maritime gray zone operations (Report No. RR-A2954-1). RAND Corporation. https://doi.org/10.7249/RRA2954-1

(b) Wasser, B., Oberholtzer, J., Pettyjohn, S. L., and Mackenzie, W. (2019). Gaming gray zone tactics: Design considerations for a structured strategic game (Report No. RR-2915-A). RAND Corporation. https://doi.org/10.7249/RR2915